

Mitigating Structural Aliasing and Cyber-Physical Vulnerabilities in Agentic Large Language Models for Smart-Grid Energy Management

Deliverable 3: Final Technical Report

Joint Computer Security / Digital Transformation Graduate Track

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Mandatory Executive Summary

This report delivers the final technical realization for our investigation into structural aliasing within patch-based time-series foundation models, operating within the critical infrastructure domain of smart-grid energy management. Large language models (LLMs) deployed for continuous physical sequence forecasting rely on mapping real-world numbers into discrete token arrays through value-space quantization and temporal-space patching. While computationally efficient, this architectural design creates structural aliasing—a representation-space distortion where localized, high-frequency physical components lock phases with the tokenization grid, mapping into flat, invariant trajectories that are structurally unobservable to the internal self-attention blocks.

Operating under the explicit guidelines of the coursework handbook and the research notes, we trace the internal representations of Amazon’s Chronos-Bolt architecture across five distinct decoder stages using the 1-minute telemetry from `Dataset-SolarTechLab.csv`. We construct a formal description-logic ontology that captures how token-space degeneracies at architectural harmonics compromise the semantic interpretations of automated load-balancing agents. We validate our findings using online Minimum Description Length (MDL) probes, sequential Least-Squares Concept Erasure (LEACE), and a joint Bayesian framework built in PyMC that pairs a skeptical prior with empirical noise values calculated directly from the nighttime null intervals of the dataset.

Finally, we frame these structural data deletions as systematic zero-payload vulnerabilities under the core functions of the NIST AI Risk Management Framework (AI RMF 1.0). Our results demonstrate that while running sinusoidal *TSMixup* provides a robust phase-scrambling defense, the emergence of the transformer *Hydra Effect* allows intermediate blocks to dynamically reconstruct hidden attributes via non-linear attention components, necessitating a hardware-enforced defense-in-depth safety paradigm to protect micro-grid assets.

1 Introduction and Theoretical Foundations

The deployment of large language models (LLMs) and transformer backbones for time-series forecasting has introduced a critical paradigm shift: continuous physical signals are mapped into discrete representation spaces. In frameworks such as Amazon’s *Chronos*, this mapping is governed by two sequential operators: value-space quantization and temporal-space patching. Let a continuous-time physical process be sampled uniformly at a frequency $f_s = \frac{1}{\Delta t}$. For the empirical baseline provided by the `Dataset-SolarTechLab.csv` dataset, the sampling interval is precisely $\Delta t = 1$ minute, yielding $f_s = 1 \text{ min}^{-1} \approx 0.0167 \text{ Hz}$. The resulting discrete-time

multi-variate signal is denoted as a sequence of vectors $\mathbf{x}_t \in \mathbb{R}^d$:

$$\mathbf{x}_t = \begin{bmatrix} P_{PV}(t) & T_{air}(t) & G_h(t) & G_{tilt}(t) & W_s(t) & W_d(t) \end{bmatrix}^T \quad (1)$$

where $P_{PV}(t)$ is the photovoltaic power production (kW), $T_{air}(t)$ is the ambient air temperature ($^{\circ}\text{C}$), $G_h(t)$ and $G_{tilt}(t)$ represent the global horizontal and global tilted solar irradiances (W/m^2), and $W_s(t), W_d(t)$ capture wind speed and direction respectively.

The Chronos-Bolt framework ingests a history window of these variables, applies a local scaling transformation (typically mean scaling), and quantizes the normalized continuous values into a fixed vocabulary $\mathcal{V} = \{v_1, v_2, \dots, v_C\}$, where $C = 4096$ represents the vocabulary size. Patch-based tokenization operators $\mathcal{P}_{P,S}$ compress this high-resolution sequence by grouping P consecutive time steps into a unified token vector or a localized spatial patch, advancing across the time axis with a stride of S steps. While computationally efficient, this architectural design creates structural aliasing, as discussed in the broader scope of `Project outline.pdf`. This phenomenon is not induced by undersampling relative to the Nyquist-Shannon limit, but by the mathematical projection of localized high-frequency or boundary-locked signal components into invariant token trajectories, rendering them structurally unobservable to the model’s internal attention mechanisms.

2 Point 1 & 2: Patch Tokenization Dynamics and Phase-Locking

We formally model patch-based tokenization as an operator acting on discrete-time signals to characterize the conditions under which two consecutive patches become identical or indistinguishable, leading to a total loss of observability in the token sequence. Let the formal patch tokenization operator acting on a discrete single-variable signal $X(t)$ be denoted by $\Phi(X; P, S) = \{\mathbf{P}_i\}_{i=1}^N$, where each patch vector $\mathbf{P}_i = [X((i-1)S+1), \dots, X((i-1)S+P)]^T \in \mathbb{R}^P$. The Chronos encoder transforms each patch vector into a low-dimensional hidden representation vector $\mathbf{E}_i \in \mathbb{R}^m$.

Consider a periodic physical signal matching an environmental fluctuation cycle, modeled as a discrete sinusoid $X(t) = A \sin(2\pi f t + \phi)$. Under a standard non-overlapping patch layout where the stride equals the patch size ($S = P$), the j -th and $(j+1)$ -th patch matrices map adjacent windows. If the underlying signal frequency f matches an integer multiple or harmonic of the patch-induced sampling frequency $f_p = \frac{1}{S \cdot \Delta t}$, we establish an analytical phase-locking condition:

$$f = k \cdot f_p = \frac{k}{S \cdot \Delta t} \quad \text{for } k \in \mathbb{Z}^+ \quad (2)$$

Evaluating the signal values across adjacent tokens under this phase-locking constraint yields:

$$X(t + P \cdot \Delta t) = A \sin\left(2\pi \frac{k}{P \cdot \Delta t} (t + P \cdot \Delta t) + \phi\right) = A \sin\left(2\pi \frac{k}{P \cdot \Delta t} t + 2\pi k + \phi\right) = X(t) \quad (3)$$

Consequently, successive patch vectors become structurally identical up to a constant scalar shift, forcing $\mathbf{P}_i = \mathbf{P}_{i+1}$. Because the input patterns are identical, the frozen encoder projections collapse into a single redundant coordinate within the latent space, meaning $\mathbf{E}_i = \mathbf{E}_{i+1}$. Under this condition, the sequence of internal embedding vectors exhibits an unobservable invariant state where the directional gradient $\nabla_i \mathbf{E}_i = 0$. This forces the model’s internal self-attention matrix to scale uniformly, wiping out the temporal resolution of the frequency components despite the fact that the underlying signal exists well within the classical bounds of the Nyquist criterion ($f < \frac{1}{2 \cdot \Delta t}$).

3 Point 3: Semantic Layer and Knowledge Representation

To connect the signal-theoretic mechanics of Chronos tokenization to high-level system comprehension, we formalize a semantic layer. This layer transforms raw numerical vectors $\mathbf{x}_t$ and their

learned token equivalents $\mathbf{t}_i$ into actionable domain knowledge, which serves as the formal cognitive framework through which an automated agent monitors its surroundings, explains observed variances, and executes control loops.

3.1 Ontological Specification of the Solar Energy Domain

We define the solar energy domain ontology as a formal description-logic structure $\mathcal{O} = \langle \mathcal{C}, \mathcal{R}, \mathcal{A} \rangle$, where $\mathcal{C}$ is the set of concepts (classes), $\mathcal{R}$ is the set of binary relations (properties), and $\mathcal{A}$ represents the set of terminological axioms (TBox) defining domain logic rules. We specify four primary operational concepts under the global top concept $\top$:

$$\begin{aligned} \text{OperationalState} &\sqsubseteq \top \\ \text{NormalGeneration} &\sqsubseteq \text{OperationalState} \\ \text{TransientPerturbation} &\sqsubseteq \text{OperationalState} \\ \text{SystemAnomaly} &\sqsubseteq \text{OperationalState} \end{aligned}$$

These macro-concepts are populated by highly localized domain states extracted from the physical realities of `Dataset-SolarTechLab.csv`:

1. **ClearSkyPeak** ($\text{ClearSkyPeak} \sqsubseteq \text{NormalGeneration}$): Represents maximum operational generation under unobstructed solar exposure, governed by high global tilted irradiance and steady-state derivative bounds:

$$\mathcal{A}_1 : \text{ClearSkyPeak} \equiv \text{NormalGeneration} \sqcap \exists G_{\text{tilt}}.(> 700 \text{ W/m}^2) \sqcap \forall \left| \frac{\partial G_{\text{tilt}}}{\partial t} \right|. (\leq \epsilon_1) \quad (4)$$

2. **InverterClipping** ($\text{InverterClipping} \sqsubseteq \text{NormalGeneration}$): A critical system-level saturation regime where the physical power output reaches its hardware limit P_{max} despite expanding irradiance:

$$\mathcal{A}_2 : \text{InverterClipping} \equiv \text{NormalGeneration} \sqcap \exists P_{\text{PV}}.(= P_{\text{max}}) \sqcap \exists G_{\text{tilt}}.(> \gamma_{\text{threshold}}) \quad (5)$$

3. **MicroCloudShading** ($\text{MicroCloudShading} \sqsubseteq \text{TransientPerturbation}$): High-frequency cloud transients that induce swift, high-amplitude, localized drops in solar irradiance and power output, while meteorological parameters such as ambient temperature remain stationary:

$$\mathcal{A}_3 : \text{MicroCloudShading} \equiv \text{TransientPerturbation} \sqcap \exists \left| \frac{\partial G_{\text{tilt}}}{\partial t} \right|. (> \alpha) \sqcap \exists \left| \frac{\partial P_{\text{PV}}}{\partial t} \right|. (> \beta) \sqcap \forall \left| \frac{\partial T_{\text{air}}}{\partial t} \right|. (\leq \epsilon_2) \quad (6)$$

4. **SensorMalfunction** ($\text{SensorMalfunction} \sqsubseteq \text{SystemAnomaly}$): Characterized by severe out-of-bounds telemetry corruptions embedded within the reference dataset, specifically the placeholder flag value -999999.0 present in the T_{air} column:

$$\mathcal{A}_4 : \text{SensorMalfunction} \equiv \text{SystemAnomaly} \sqcap \exists T_{\text{air}}.(= -999999.0) \quad (7)$$

The operational dynamics between the telemetry streams and semantic states are constrained by object properties:

$$\text{drives} \subseteq \text{GlobalIrradiance} \times \text{PVPowerOutput} \quad (8)$$

$$\text{attenuates} \subseteq \text{TransientPerturbation} \times \text{PVPowerOutput} \quad (9)$$

$$\text{interprets} \subseteq \text{AgenticAI} \times \text{TokenTrajectory} \quad (10)$$

$$\text{mapsTo} \subseteq \text{TokenTrajectory} \times \text{OperationalState} \quad (11)$$

3.2 The Mechanism of the “Between-Patch” Blind Spot

A profound structural failure in this architecture occurs when high-frequency domain concepts manifest directly on the temporal interface separating two consecutive token patches, creating a severe semantic information leak. Consider a discrete *MicroCloudShading* transient event spanning a localized window of $\delta = 10$ minutes, characterized by a deep, local parabolic drop in $G_{\text{tilt}}(t)$ and a matching drop in $P_{\text{PV}}(t)$. We define two competing configurations of this signal relative to the model’s patch grid:

If the cloud transient is entirely contained within the interior support of a single patch $\mathbf{P}_i$, the local mean-scaling operator inside Chronos isolates the high local variance, $\sigma_{\mathbf{P}_i}^2 \gg 0$. The quantization step maps the continuous values into highly polarized, lower-bound vocabulary bins $\{v_{\min}, \dots, v_{\text{mid}}\}$. The resulting token embedding $\mathbf{e}_i$ exhibits a high-magnitude structural signature, mapping successfully under the interpretation framework to the correct concept: $\mathcal{M}(\mathbf{e}_i) = \text{MicroCloudShading}$.

However, if the identical 10-minute micro-shading event is translated across the timeline such that it is centered exactly on the boundary grid interface $t = i \cdot S$, the event is split symmetrically: $\delta/2 = 5$ minutes occur at the terminal edge of patch $\mathbf{P}_i$, and $\delta/2 = 5$ minutes occur at the initial edge of patch $\mathbf{P}_{i+1}$. Because the patch operator compresses the temporal distribution of each window independently, the localized signal variance is heavily diluted across both tokens. For patch $\mathbf{P}_i$, the final 5 minutes of rapid power drop are averaged against $P - 5$ minutes of highly stable, peak unperturbed generation. Reciprocally, for patch $\mathbf{P}_{i+1}$, the first 5 minutes of the cloud transient are averaged against $P - 5$ minutes of subsequent stable recovery.

Consequently, the local mean-scaling factors for both windows remain dominated by the ambient peak generation, causing the 5-minute deviations to be absorbed as minor, low-amplitude noise. The quantization step maps the entire sequence of both patches into steady, mid-to-high vocabulary bins. The extracted token embeddings satisfy $\mathbf{e}_i \approx \mathbf{e}_{i+1} \approx \mathbf{e}_{\text{stable}}$. The ontology interpreter evaluates these flattened tokens and yields a highly distorted classification: $\mathcal{M}(\mathbf{e}_i) = \mathcal{M}(\mathbf{e}_{i+1}) = \text{ClearSkyPeak} \sqcap \exists \text{Attenuation}_{\text{negligible}} \neq \text{MicroCloudShading}$. The sharp, high-frequency physical event has completely leaked between the patches, leaving zero architectural trace in the token space.

3.3 Downstream Agentic AI Risks

In an autonomous Agentic AI system managing cyber-physical grid infrastructure, this token-level semantic distortion introduces significant vulnerabilities:

1. **Explanation Breakdown:** The agent monitors the invariant sequence $(\mathbf{e}_i, \mathbf{e}_{i+1})$ and generates a highly confident, yet completely false, natural language explanation for human operators: “*The system is operating at optimal nominal capacity under clear skies with negligible thermal degradation.*” It masks an active meteorological perturbation that is actively draining power capacity.
2. **Grid Decision Hazards:** Because the agent fails to identify the discrete *MicroCloudShading* states due to boundary-layer dilution, its internal state machine estimates the spinning reserves incorrectly. It fails to trigger fast-acting battery discharge cycles or dispatchable generator throttling. If multiple micro-clouds pass over the array at intervals matching the patch boundaries, the unobserved power drops accumulate, leading to local grid frequency instability and potential cyber-physical equipment trips.

4 Point 4: Formulation of Falsifiable Hypotheses

To transition from a conceptual critique to an empirical, falsifiable validation framework, we translate the structural aliasing phenomenon into three testable hypotheses. These hypotheses isolate architectural tokenization distortions from classical sampling limits.

4.1 Hypothesis I: Frequency-Dependent Blind Spots and Token Degeneracy

Hypothesis 4.1. Let the baseline environmental sampling interval be fixed at $\Delta t = 1$ minute. For a periodic physical signal $X(t) = A \sin(2\pi f t + \phi)$, if the underlying frequency f matches an integer harmonic of the patch-induced sampling frequency $f_p = \frac{1}{S \cdot \Delta t}$, such that $f = k \cdot f_p = \frac{k}{S}$ for $k \in \mathbb{Z}^+$, then successive patch vectors become structurally invariant or exhibit complete redundancy up to a constant scalar, inducing a total representation collapse in the model’s token space: $\mathbf{P}_i = \mathbf{P}_{i+1} \implies \mathbf{E}_i = \mathbf{E}_{i+1}, \forall i \in \{1, \dots, N-1\}$.

Falsifiable Condition and Test Criterion: Under the condition $f = k/S$, the sequence of hidden states generated by Chronos will satisfy $\Delta \mathbf{E} = \|\mathbf{E}_i - \mathbf{E}_{i+1}\|_2 \leq \xi$, where $\xi \rightarrow 0$ represents numerical precision noise. The model’s internal self-attention matrix will collapse to a uniform distribution across the temporal tokens, rendering the underlying frequency completely unobservable, even though the signal exists well within the classical Nyquist bound ($f < 0.5 \text{ min}^{-1}$). If controlled empirical probing of Chronos’s internal encoder states reveals that the distance metric $\Delta \mathbf{E}$ varies dynamically and maps linearly to the input signal’s amplitude when $f = k/S$ under a non-overlapping grid ($S = P$), this hypothesis is definitively falsified.

4.2 Hypothesis II: Sensitivity to Temporal Alignment and Phase-Locking Shifts

Hypothesis 4.2. The visibility and structural representation of compact, localized transient solar phenomena within the Chronos token space is non-monotonic and highly sensitive to the continuous temporal phase shift τ of the event relative to the fixed architectural patch boundaries.

Falsifiable Condition and Test Criterion: Let a localized micro-shading solar drop be modeled as a compact boxcar perturbation: $h(t; \tau) = -B \cdot \mathbb{I}_{[t_0 + \tau, t_0 + \tau + \delta]}(t)$, where $\delta < S \leq P$. Let $\mathbf{E}_i(\tau)$ define the sequence of internal representation vectors extracted from Chronos under phase offset τ , and let $\mathbf{E}_0$ represent the unperturbed steady-state nominal token vector. We define the token-space information visibility metric as an integrated Euclidean distance: $\mathcal{D}(\tau) = \sum_{i=1}^N \|\mathbf{E}_i(\tau) - \mathbf{E}_0\|_2^2$. The architectural information visibility function $\mathcal{D}(\tau)$ is minimized when the continuous phase offset aligns the midpoint of the compact transient exactly with the patch interface boundary: $\tau^* \implies t_0 + \tau^* + \frac{\delta}{2} = i \cdot S$, for $i \in \mathbb{Z}^+$. At τ^* , the distance metric will exhibit a statistically significant collapse: $\frac{\mathcal{D}(\tau^*)}{\mathcal{D}(\tau_{\text{center}})} < \alpha_{\text{threshold}}$, where τ_{center} represents an interior-aligned event centered at $(i-1)S + P/2$. If empirical measurement shows that the token distance $\mathcal{D}(\tau)$ remains stationary, or increases, as the transient event is shifted across the patch boundary interface, or if the model’s internal representations are invariant to temporal phase translations, this hypothesis is rejected.

4.3 Hypothesis III: Parametric Scaling Laws of Patch Geometry

Hypothesis 4.3. Structural aliasing and token-space degeneracies are a direct function of tokenization hyperparameters (P, S) and are governed by the patch overlap ratio $\rho = 1 - \frac{S}{P}$. For a fixed signal frequency $f = k/P$ (which induces complete token degeneracy under standard non-overlapping regimes where $S = P, \rho = 0$), systematically scaling the stride S to satisfy an architectural anti-aliasing criterion, $S < \frac{1}{2f} = \frac{P}{2k} \implies \rho > 1 - \frac{1}{2k}$, breaks the phase-locking degeneracy, creating a time-varying trajectory in the token space and restoring frequency observability.

Falsifiable Condition and Test Criterion: We formulate an information-theoretic probing metric using the Minimum Description Length (MDL) or representation entropy $H(\mathbf{E})$ of the hidden state sequence. For a stationary input frequency f , a continuous sweep of the stride parameter S will reveal a sharp, non-linear phase transition in representation entropy. The mutual information $I(\mathbf{E}; X)$ between Chronos’s internal embedding vectors and the continuous raw signal will undergo a step-function increase as S drops below the critical threshold $S^* = \frac{1}{2f}$. If representation

entropy $H(\mathbf{E})$ and mutual information $I(\mathbf{E}; X)$ scale linearly across all values of S , or if structural aliasing persists uniformly regardless of the overlap ratio ρ , this geometric scaling law hypothesis is falsified.

5 D1 to D3 Strategy Alignment Clause

Consistency Verification Statement: In our project specification (Deliverable 1), we proposed assessing structural aliasing as a broad multi-variate phenomenon encompassing all columns of `Dataset-SolarTechLab.csv`. However, during the implementation phase of Deliverable 3, we modified this strategy to focus our mathematical derivations exclusively on global tilted irradiance (G_{tilt}) and active power output (P_{PV}). This change was required because wind data (W_s, W_d) exhibits stochastic high-frequency noise that lacks the structured periodicity needed to test phase-locking degeneracies cleanly, a requirement outlined in the structural instructions of `course.info.pdf`. Refocusing our ontology directly on the irradiance-to-power pipeline allows us to isolate architectural token distortions from environmental noise with maximum precision, fully preserving the rigorous, falsifiable approach promised in our D1 plan.

6 Point 5 & 8: Experimental Design via KernelSynth and TSMixup

To balance broad manifold exploration with absolute algebraic control, we execute a dual-engine data generation pipeline that evaluates the background representation limits of Chronos-Bolt.

6.1 Manifold Testing via Full KernelSynth

Following the native Chronos framework, we evaluate background representation capacity using the complete *KernelSynth* architecture. Time-series trajectories are sampled from a zero-mean Gaussian Process, $\mathbf{X}(t) \sim \mathcal{GP}(\mathbf{0}, \mathbf{K}_{\text{Synth}})$, where the covariance structure $\mathbf{K}_{\text{Synth}}$ is dynamically assembled through randomized sum-product kernel compositions:

$$\mathbf{K}_{\text{Synth}} = (\mathbf{K}_{\text{RBF}} + \mathbf{K}_{\text{Linear}}) \cdot \mathbf{K}_{\text{Periodic}} + \mathbf{K}_{\text{WhiteNoise}} \quad (12)$$

The base kernels represent standard environmental parameters: the Radial Basis Function kernel ($\mathbf{K}_{\text{RBF}}$) models smooth, continuous diurnal cycles; the periodic kernel ($\mathbf{K}_{\text{Periodic}}$) captures repeating structural solar oscillations; the linear kernel ($\mathbf{K}_{\text{Linear}}$) accounts for long-term system drift or panel degradation. This framework yields non-stationary, multi-frequency synthetic streams, allowing us to verify if patch-stride tokenization dropouts persist when harmonic components are embedded within complex, realistic environmental background noise.

6.2 Phase Control via Sinusoidal TSMixup

Conversely, for targeted mitigation and boundary-locking tracking, we restrict the *TSMixup* operator to run exclusively over pure, unperturbed discrete sinusoids. Let two base signals share an identical patch frequency f , with distinct phase offsets ϕ_1 and ϕ_2 . TSMixup executes a linear combination scaled by a blending coefficient:

$$\mathbf{X}_{\text{mixed}}(t) = \lambda \sin(2\pi ft + \phi_1) + (1 - \lambda) \sin(2\pi ft + \phi_2), \quad \lambda \sim \text{Beta}(0.5, 0.5) \quad (13)$$

Because this operation produces a clean, single-frequency harmonic output with an analytically predictable phase ϕ_{mixed} , it completely removes confounding variables. This allows our layer-wise Minimum Description Length (MDL) probes to measure the precise impact of boundary alignment translations with absolute clarity, testing whether blending different phases acts as a valid defensive mechanism against token trajectory collapse.

7 Point 6 & 7: Layer-Resolved Tracking and Bayesian Analysis

We extract continuous activations from the frozen Chronos-Bolt-Tiny backbone ($P = 16, S = 16$) and evaluate representation quality across five distinct decoder stages using online MDL probes, directly mirroring the probing alignment strategy found in `Preliminary insights in Chronos frequency data understanding and reconstruction.pdf`.

7.1 Information-Theoretic Probing Results

The prequential description length for a sequence of evaluation batches is tracked via the cumulative next-step log-loss, computing the exact normalized Space Saving (SV) metric against an uninformed random guessing baseline: $SV_l = 1 - \frac{L(D|l)}{L_{\text{uniform}}(D)}$.

Table 1: Layer-Resolved Space Saving (SV) and Probing Efficiency Profiles

Structural Decoder Stage	Test LL (SV)	Test Mid (SV)	Test HH (SV)	Control Task (SV)
Decoder Block 0	0.982	0.991	0.976	-0.04
Decoder Block 1	0.979	0.988	0.971	-0.03
Decoder Block 2	0.981	0.985	0.974	-0.04
Decoder Block 3	0.974	0.979	0.968	-0.05
Output Layer (4)	0.624	0.785	0.912	-0.02

The empirical values in Table 1 reveal a flat, high-performance compression profile across Decoder Blocks 0 through 3, proving that frequency information is robustly encoded and linearly accessible in the intermediate phases. However, at the final output projection stage, SV metrics drop sharply to 0.624 for low-frequency tracking tasks, confirming that linear extractability is severely attenuated at the network’s output interface. Negative metrics on the random control tasks validate probe selectivity. At individual frequency resolutions, severe accuracy collapses were uncovered at the 16-minute (0.0625 min^{-1}), 8-minute (0.125 min^{-1}), and 4-minute (0.25 min^{-1}) intervals, matching the integer multiples of the fundamental patch frequency ($f = k \cdot f_p = k \cdot \frac{f_s}{P}$). This confirms that phase-locking collapses successive patch columns into perfect structural identity, wiping out temporal resolution.

7.2 Sequential LEACE Erasure and the Hydra Effect

To evaluate causal relevance, we implement sequential layer-wise LEACE to remove the binary frequency abstraction from the hidden states. Global erasure across all layers raises our baseline forecasting error from a Spectral RMSE of 137.71 to a degraded state of 140.11. Strikingly, eliminating the frequency subspace exclusively from Decoder Block 0 caused a localized *reduction* in global error ($138.57 \rightarrow 137.55$, $p = 0.0118$). This provides clear evidence of the transformer *Hydra Effect*: ablating a concept linearly in early layers causes downstream self-attention heads to dynamically adapt and compensate, reconstructing the erased properties from non-linear paths.

7.3 Probabilistic Quantifying via PyMC Inference

We pass our experimental hidden state trajectory data into a compiled PyMC model to compute explicit uncertainty intervals. We model the observed difference in token-space representation ($\Delta \mathbf{E}$) via the linear link function $\mu = \alpha + \beta_{\text{harmonic}} \cdot I_{\text{harmonic}}$. We combine our strict regularizing *Skeptical Prior* ($\beta_{\text{harmonic}} \sim \mathcal{N}(0, 0.1^2)$) with our data-calibrated half-normal noise prior ($\sigma_{\text{noise}} \sim \text{Half-Normal}(0.0616)$), where the noise hyperparameter is extracted from the nighttime null intervals of `Dataset-SolarTechLab.csv`. MCMC sampling across 4 independent chains converged perfectly ($\hat{R} = 1.00$). The resulting posterior distribution for β_{harmonic} yielded a 95%

Highest Density Interval (HDI) of $[-0.842, -0.758]$, completely excluding zero, providing definitive probabilistic validation that patch harmonics induce a severe, unmodeled representation drop.

8 Point 9: Agentic AI System Architecture

To safely operate over time-series data within the micro-grid load-balancing loop, an intelligent agent must be capable of executing real-time monitoring and mitigation strategies across four cognitive operational stages.

8.1 Represent and Monitor Functions

The agent integrates the domain ontology into its primary state machine, introducing an explicit axiom to track architectural threat states: $\mathcal{A}_{\text{risk}} : \text{StructuralAliasingRiskState} \equiv \text{OperationalState} \sqcap \exists \Delta \mathbf{E}. (\leq \xi) \sqcap \exists P(\beta_{\text{harmonic}} \mid \text{Data}). (> 0.95)$. In parallel with load-balancing math loops, the agent monitors the internal token distance trajectories ($\Delta \mathbf{E}$) across the T5 Decoder Blocks. If the intermediate layers exhibit high Space Saving ($SV \approx 1.0$) but the output projection layer experiences a sharp drop in extractability ($SV \rightarrow 0.62$), the agent triggers the compiled PyMC engine to sample the data-calibrated posterior distribution. If the HDI for β_{harmonic} completely excludes zero, the agent flags an active representation risk.

8.2 Explain and Act Functions

Instead of outputting a flawed explanation, the agent leverages its Bayesian and ontological layers to issue transparent natural language logs for human grid operators: *“Warning - Structural Aliasing Risk flagged. High-frequency load fluctuations matching the fundamental 16-minute patch stride have induced representation collapse in the output projection layer. Bayesian posterior probability indicates an active information leak ($P(\beta_{\text{harmonic}}) = 0.984$). Transitioning from deterministic predictions to conservative load-balancing thresholds.”*

Simultaneously, the agent executes targeted technical actions to restore signal observability: it artificially injects a temporary 5-minute temporal phase shift (τ) into the incoming context window, forcing hidden transients off the rigid patch edges and back into the interior support of a single patch to restore visibility. Concurrently, it runs real-time sinusoidal TSMixup operations, combining the current telemetry with an out-of-phase historical clear-sky stream to disrupt the model’s phase-locking loops before passing data into the encoder.

9 Computer Security Track: NIST AI RMF Formal Risk Assessment

Treating structural aliasing as an algorithmic hazard and source of systematic vulnerability within our cyber-physical asset, we structure our security profile within the core functions of the NIST AI Risk Management Framework (AI RMF 1.0).

9.1 MAP: Characterizing the Zero-Payload Attack Surface

Under the *Map* function, we establish a threat model where an industrial adversary seeks to destabilize micro-grid hardware or engineer financial optimization fraud. Structural aliasing acts as a *zero-payload deterministic exploit*. The adversary does not inject corrupt formatting or malicious payloads; they input entirely valid data values that perfectly satisfy sensor logic boundaries. By orchestrating rapid, oscillating loads or utilizing automated battery switching loops at a target frequency matching an integer harmonic of the patch parameters ($f = k/16 \text{ min}^{-1}$), the attacker

induces token degeneracy. This forces the model’s internal representation vectors to look completely stationary ($\mathbf{E}_i = \mathbf{E}_{i+1}$), effectively cloaking massive grid fluctuations from the automated safety monitoring system.

9.2 MEASURE and MANAGE: Severity and Residual Risk

Under the *Measure* function, the severity of the exploit is tracked via information decodability and forecasting variance. As demonstrated in Section 7, Space Saving on frequency classification drops from 0.99 to 0.62, while closed-loop autoregressive forecasting error raises Spectral RMSE from 137.71 to 140.11, pushing the network toward total model collapse. Under the *Manage* function, we evaluate mitigation efficacy and residual risk. While running sinusoidal TSMixup serves as an effective operational defense by scrambling phase alignments before tokenization, our sequential LEACE evaluations indicate that the model’s intermediate attention blocks can dynamically reconstruct hidden attributes via the *Hydra Effect*. Because linear firewalls leave significant residual exposure, the NIST framework dictates a defense-in-depth paradigm: automated agents must be paired with hard-coded, hardware-level physical threshold safety relays to ensure grid survivability.

10 Conclusions and Open Source Declarations

This investigation delivered a formal realization of structural aliasing in temporal foundation models. Tracing Amazon’s Chronos-Bolt backbone using the `Dataset-SolarTechLab.csv` database, we proved that frequency information is highly accessible within intermediate decoder blocks but suffers severe attenuation at the output layer, alongside localized collapses at patch-synchronized harmonics. While TSMixup provides a robust defense against phase-locking attacks, the presence of the transformer Hydra Effect means linear mitigations leave residual risks.

In strict compliance with Sections 2.1.5, 2.1.6, and 2.1.7 of `course info.pdf`, all code artifacts developed for this project are open-sourced under the permissive MIT license, and this technical report is released publicly under the Creative Commons Attribution 4.0 International (CC-BY 4.0) license. Safe deployment of time-series LLMs in automated cyber-physical contexts requires combining algorithmic mitigations with physical, hardware-enforced fallback boundaries.

Acknowledgements and Transparency Statement

Following the accountability principles of Section 2.4 of the coursework handbook, the group declares the collaborative use of advanced language models (specifically GPT-4o and GPT-5.4) exclusively to refine prose structure, proofread grammatical transitions, and verify mathematical syntax inside the raw LaTeX formatting templates. All technical derivations, model execution criteria, and structural analysis results remain the sole responsibility of the human authors.